

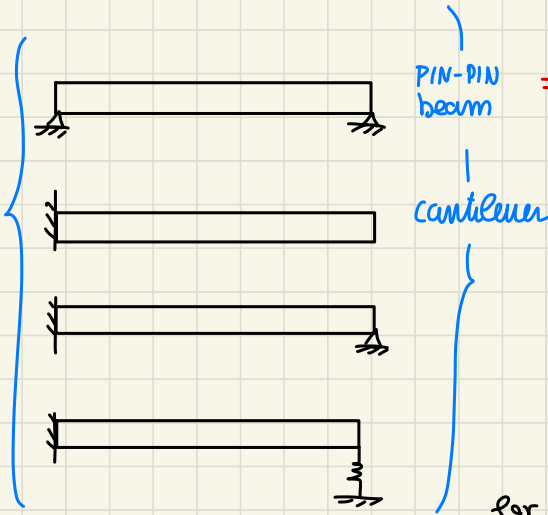
ESE 16 VIBRATION of DPS (Beams)

using stationary solution for different beams motion

→ FOCUS on **Bending vibrations** (most meaningful components of vibration)
(more complex mathematically) → because involve $\sin, \cos, \sinh, \dots$

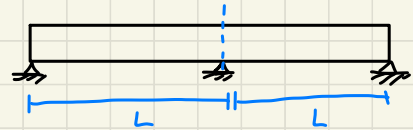
FINITE ELEH. Method → applied on more application cases
Geometry simpler until now, but useful!

ES) until now single beam constrained



new **ES**

including a 3RD PIN
on center, so additional
constraints on middle



2L overall,
pin far as L in between

for **BENDING VIBRATIONS** STATIONARY SOLUTION

$$(X) W(x,t) = [A \cos(\delta x) + B \sin(\delta x) + C \cosh(\delta x) + D \sinh(\delta x)] \cos(\omega t + \phi)$$

NOT ↑ we can't use this solution! → when found we assumed that

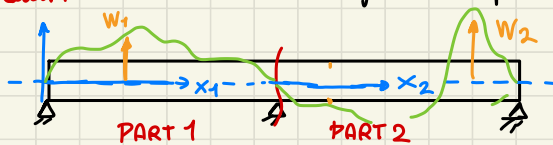
CORRECT SOLUTION to rep. entire syst vibration

constraint forces at the ends

No forces where acting inside the L beam but only @ $x=0, x=L$

now new constraint force in the middle

⇒ IF we divide the syst in 2 pieces



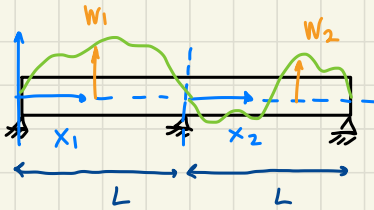
$\left\{ \begin{array}{l} x_1 \text{ for PART 1} \\ x_2 \text{ for PART 2} \end{array} \right\}$ we introduce 2 solutions W_1, W_2
PART 1 PART 2

separately they full fill the assumption

we address solution by 2 coordinates

x_1, x_2 along a beam axis

each parts respect the assumption for which
 we found $W(x,t) \rightarrow$ so
 $W_1(x_1,t)$, $W_2(x_2,t)$ full fill



$$W_1(x_1,t) = [A_1 \cos(\delta x_1) + B_1 \sin(\delta x_1) + C_1 \cosh(\delta x_1) + D_1 \sinh(\delta x_1)] \cos(\omega t + \varphi)$$

$$W_2(x_2,t) = [A_2 \cos(\delta x_2) + B_2 \sin(\delta x_2) + C_2 \cosh(\delta x_2) + D_2 \sinh(\delta x_2)] \cos(\omega t + \varphi)$$

obviously also constants are different, we expect different
 right - left part $\Rightarrow A, B, C, D$ take different values on W_1, W_2
 (1), (2) coefficients

UNKNOWN $\Downarrow$

$$\underline{x} = (A_1 \ B_1 \ C_1 \ D_1 \ A_2 \ B_2 \ C_2 \ D_2)^T$$

respect ω on the two parts? $\rightarrow$ (NO) the natural freq. is a property
 of entire system, CANNOT be different between 2 PARTS

(or system parts moving differently, NOT meaningful!) $\rightarrow$ same ω

{ if $\omega_1 \neq \omega_2$ we cannot guarantee same displacement @ $x = L$ }

also the phase $\varphi_1 = \varphi_2 = \varphi$ function of time should be same

$\rightarrow$ on $x = L$ where attached, they must behave equally!

γ values? $\rightarrow$ because

$$\gamma^4 = \frac{m \omega^2}{EI}$$

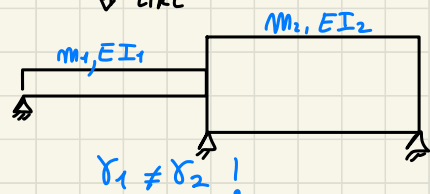
M/L
 bending stiffness

We can
 address
 solution

• If PART 1 and PART 2
 has same m, EI
 then $\gamma_1 = \gamma_2 = \gamma$

$\leftarrow$ we
 work
 on this
 condition

• BUT of different
 m, EI $\gamma_1 \neq \gamma_2$
 $\downarrow$ LIKE



$\gamma_1 \neq \gamma_2 !$

WHEN addressing one piece of beam $\rightarrow$ 4 boundaries condition where enough
 $2 \text{ \& } \text{end } x=0, x=L$

4 boundaries allow to

write 4 eq lin omogeneous in 4 unk as $[H(xL)] \underline{x} = 0$ where $\underline{x} \in \mathbb{R}^4$

NOT TRIVIAL $\downarrow$ $\det(H(xL)) = 0 \rightarrow \delta_k \quad k=1,2,\dots$

$\forall \delta_k$ extract $\underline{x}_k \Leftarrow \omega_k \quad k=1,2,\dots$

NONTRIVIAL $\rightarrow \phi_k(x)$ MODAL SHAPE

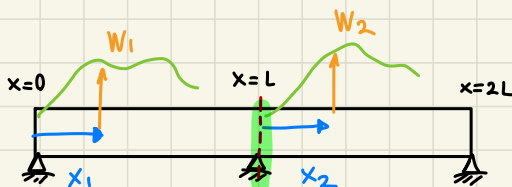
waveform of deformation $\forall \omega_k$

$\downarrow$ NOW We formulate equally the problem, with same numerical method, BUT NOW

$\underline{x} \in \mathbb{R}^8 \rightarrow$ we need $[H(xL)]_{8 \times 8}$

$\rightarrow$ 8 lin omog eq. in 8 unk

$\rightarrow$ establish for entire syst 1+2 totally 8 boundaries condition



ON SIMPLY SUPPORT beam

Q displacem, Q bending moment $\forall x=0, L$

$$M = EI \frac{\partial^2 W}{\partial x^2}$$

@ $x=0$
 ①, ②

PIN @ $x=0$
 RELATED to W_1 of PART 1

$$\begin{cases} W_1|_0 = 0 & \textcircled{1} \\ \frac{\partial^2 W_1}{\partial x_1^2}|_0 = 0 & \textcircled{2} \end{cases}$$

PIN @ $x=2L$
 W_2

$$\begin{cases} W_2|_L = 0 & \textcircled{7} \\ \frac{\partial^2 W_2}{\partial x_2^2}|_L = 0 & \textcircled{8} \end{cases}$$

$\uparrow x_2=L$ @ right end

We search here the other 4 condition, math separation between W_1, W_2 BOUNDARIES sections of JOINING, for general $A, B, \dots$ NO GUARANTEE until now that @ $x=L$ are same $\rightarrow$ we must set condition

$$W_1|_L = W_2|_0 \Rightarrow W_1|_L - W_2|_0 = 0 \quad \textcircled{3} \quad \text{homogeneous eq}$$

rotation of section related to transversal displacement gradient PART 1 and (2)

should have same ROTATION We need to specify NOT relative rotation (1), (2) CLUMPED together!

$$\frac{\partial W_1}{\partial x_1}|_L - \frac{\partial W_2}{\partial x_2}|_0 = 0 \quad \textcircled{4} \quad \text{same ROTATION}$$

then, having a PIN @ center $x_1=L \rightarrow$ here we should have $w_1=w_2=0$
 one of w_1 or w_2 should be fixed to 0, it is the same $w_1|_L=0$

better using $x_2=0 \rightarrow w_2|_0=0$
 Which simplify the expression

$$w_2|_0=0 \quad (5)$$

IF Writing
 (6) $w_1|_L=0$ is redundant, it's the same of (5) and (3) together
 any δ value satisfy $\det(H)=0$ if Writing this, because I hold a
 NON SING MATRIX $\rightarrow$ CAREFULL!

We still need a new boundary condition

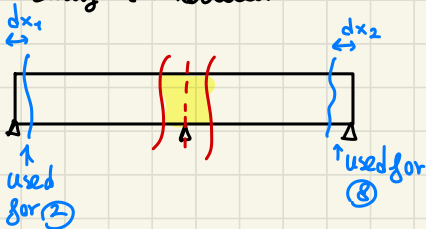
Respect KINEMATIC Relationship (displace/rotation...) all DONE

$\Rightarrow$ IT remain BALANCE of forces to consider

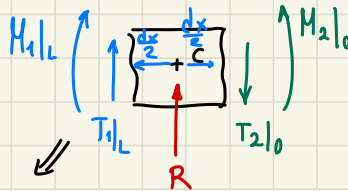
EQUIVALENT BEND MOMENT

($\uparrow$ $\boxed{+}$ $\downarrow$) SIGN CONVENTIONS

Cutting a dx between PART 1 vs 2



Consider equilibrium of central part



2 eq

Equations D'Alemb.

(PIN REACTION FORCE)

• inertia negligible HIGH ORDER fixed $\ddot{x} \approx 0$

~~$\sum F_y = 0$~~ $\rightarrow$ including R UNK

$\sum M_c = 0$ C geom center of small piece

theam

$\frac{dx}{2}$ of T_1, T_2 is almost negligible $\Rightarrow$

$$M_1|_L = M_2|_0$$

$$EI \frac{\partial^2 w_1}{\partial x_1^2} \Big|_L = EI \frac{\partial^2 w_2}{\partial x_2^2} \Big|_0$$

$$(6) \quad \frac{\partial^2 w_1}{\partial x_1^2} \Big|_L - \frac{\partial^2 w_2}{\partial x_2^2} \Big|_0 = 0$$

found all boundaries condition, we should write it as equations

$$W_i(x_i, t) = [A_i \cos(\gamma x_i) + B_i \sin(\gamma x_i) + C_i \cosh(\gamma x_i) + D_i \sinh(\gamma x_i)] \cos(\omega t + \varphi)$$

$$\frac{\partial W_i}{\partial x_i} = \gamma [-A_i \sin(\gamma x_i) + B_i \cos(\gamma x_i) + C_i \sinh(\gamma x_i) + D_i \cosh(\gamma x_i)] \cos(\omega t + \varphi)$$

$$\frac{\partial^2 W_i}{\partial x_i^2} = \gamma^2 [-A_i \cos(\gamma x_i) - B_i \sin(\gamma x_i) + C_i \cosh(\gamma x_i) + D_i \sinh(\gamma x_i)] \cos(\omega t + \varphi)$$

$\downarrow$ $\cos(\omega t + \varphi) \forall t$ valid, simplified $\downarrow$

$$\left\{ \begin{array}{l} \textcircled{1} A_1 + C_1 = 0 \\ \textcircled{2} -A_1 + C_1 = 0 \\ \textcircled{3} A_1 \cos(\gamma L) + B_1 \sin(\gamma L) + C_1 \cosh(\gamma L) + D_1 \sinh(\gamma L) - A_2 - C_2 = 0 \\ \textcircled{4} -A_1 \sin(\gamma L) + B_1 \cos(\gamma L) + C_1 \sinh(\gamma L) + D_1 \cosh(\gamma L) - B_2 - D_2 = 0 \\ \textcircled{5} A_2 + C_2 = 0 \\ \textcircled{6} -A_1 \cos(\gamma L) - B_1 \sin(\gamma L) + C_1 \cosh(\gamma L) + D_1 \sinh(\gamma L) + A_2 - C_2 = 0 \\ \textcircled{7} A_2 \cos(\gamma L) + B_2 \sin(\gamma L) + C_2 \cosh(\gamma L) + D_2 \sinh(\gamma L) = 0 \\ \textcircled{8} -A_2 \cos(\gamma L) - B_2 \sin(\gamma L) + C_2 \cosh(\gamma L) + D_2 \sinh(\gamma L) = 0 \end{array} \right.$$

$\uparrow$ LIN SYSTEM in 8 homog. equations in 8 unkwn $\rightarrow$ H?

$$[H(\gamma L)] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \cos(\gamma L) & \sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) & -1 & 0 & -1 & 0 \\ -\sin(\gamma L) & \cos(\gamma L) & \sinh(\gamma L) & \cosh(\gamma L) & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & \cos(\gamma L) & \sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ 0 & 0 & 0 & 0 & -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \end{bmatrix} \begin{array}{l} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \textcircled{4} \\ \textcircled{5} \\ \textcircled{6} \\ \textcircled{7} \\ \textcircled{8} \end{array}$$

so we found the matrix that can be managed by the script

on the code we should manage also more piece of beam...

so instead of 4×4 matrix $\rightarrow 8 \times 8$ matrix, so check for multiple $(4 \times m) \times (4 \times m)$ matrix

+ function Implementing $[H(rL)]$ on **MATLAB**

$\downarrow$ "continuous beam"

namefunc = 'cont-beam' $\gg$ Vib-dps_2022117.m

some SET of DATA of previous example...

here $L=2$ is semi-length $L_{TOT} = 4m$ overall of two pieces

cont-beam match the found $[H(rL)]$ matrix, by hand

dormans: 2 (portion on which dividing the system)

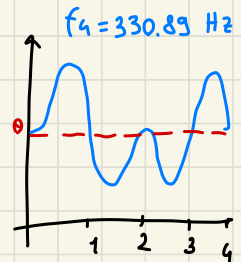
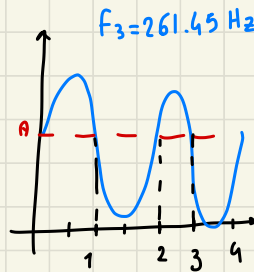
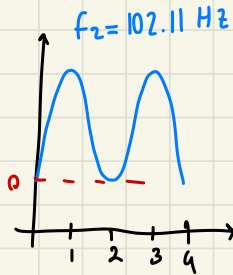
$\hookrightarrow$ because $8 \times 8 \rightarrow$ divide in 2

$f_{MAX} ? = 400 \text{ Hz}$

$\downarrow$ we find 4 nat freq on this range



0 displacement
as fixed on
① ⑤ ⑨ constrain



modal shapes are perfectly antisymm @ f_1 respect half left-right

or perfectly symm @ f_2 : equal @ left-right

@ f_3 also perfect antisymm

@ f_4 perfectly symm

$\rightarrow$ this pattern is
followed also @ higher freq

$\downarrow$

SYSTEM Geometrically
/ structurally symmetric



uniform material / property

same syst
is flipped
respect this axis !

$\Rightarrow$ we expect $\Rightarrow$ perfectly symmm / anti-symmm shape

Curvature in central PIN...

The half modal shape is equal to single PIN-PIN beam.

MIRRORED antisymmm @ second part

f_1 expected as the same of the mm mat. freq
of a single PIN-PIN beam
having $L = 1/2 L_{tot}$

$$f_1 = 65.36 \quad L = 2m$$

equally f_3 is like shape of 2nd mode of PIN-PIN beam
singular, symmm reflected on 2nd beam half

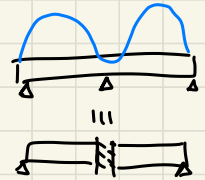
$$f_2 = 261.45Hz \quad L = 2m$$

or simply
supp beam

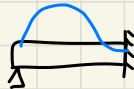
instead

f_2, f_4 SYMM, same shape of reflecting around $y=2$

found of cue fix NO ROTATION in the center of beam



it is like having a
clamp @ $x=L$



prevent displace
and rotation

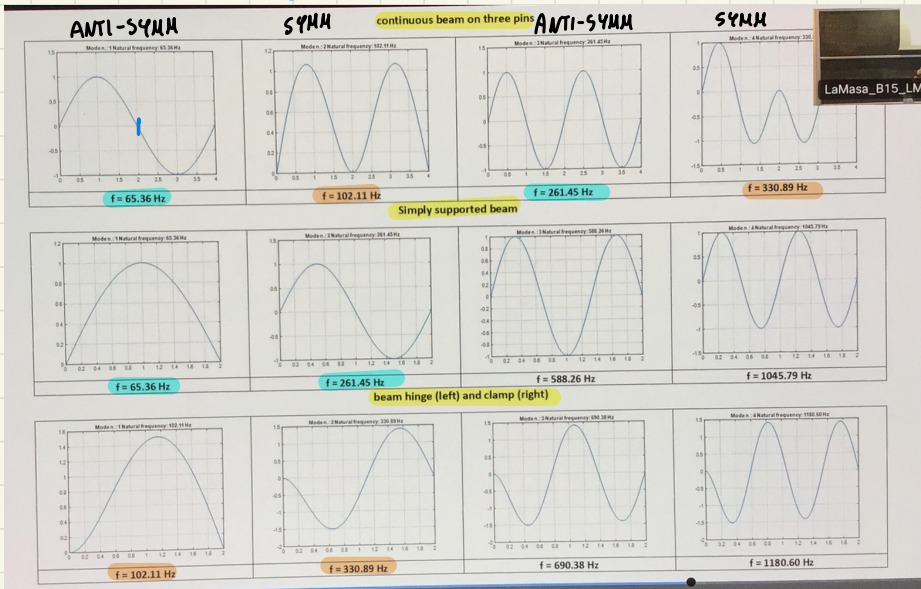
$$f_2 = f_1 \text{ of CLAMP-PIN beam}$$

equals. to same example $L = 2m$ BEAM

$$f_4 = f_2 \text{ of CLAMP-PIN beam}$$

We expect this $\nearrow$
behaviour!

RESULT of this comparison



as expected from previous resonance

Typical effect of Symm Systems

ES2

uniform section - material property beam

easy to see that on this new case

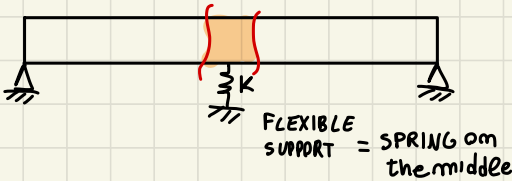
RESPECT ES1

①, ②, ⑦, ⑧

at the ends remain the same

BUT also ③ ④ ⑥ remain valid

↑ still same $M_1 = M_2$



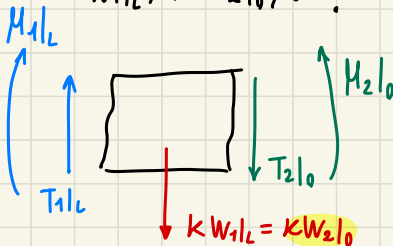
BUT ~~⑤~~ no more valid

flexible spring: meaning that

$$W_1|_L \neq 0 \quad W_2|_0 \neq 0 !$$



from equilibrium of small section



⑤ being $T = EI \frac{\partial^3 W}{\partial x^3}$

$$T_1|_L - T_2|_0 - K W_2|_0 = 0$$

$$EI \frac{\partial^3 W_1}{\partial x_1^3} \Big|_L - EI \frac{\partial^3 W_2}{\partial x_2^3} \Big|_0 - K W_2|_0 = 0$$

directed = to F $\uparrow$ (+) elongation

↑ easier equation

same system of equation, except for ⑤

being $\frac{\partial^3 w_i}{\partial x_i^3} = \gamma^3 [A_i \sin(\gamma x_i) - B_i \cos(\gamma x_i) + C_i \sinh(\gamma x_i) + D_i \cosh(\gamma x_i)] \cos(\omega t + \varphi)$

⑤

$$\cancel{EI\gamma^3} [A_1 \sin(\gamma L) - B_1 \cos(\gamma L) + C_1 \sinh(\gamma L) + D_1 \cosh(\gamma L)] - \cancel{EI\gamma^3} [-B_2 + D_2] - \underbrace{k}_{EI\gamma^3} [A_2 + C_2] = 0$$

so we can rewrite respect ES1

ROW 5 of $[H(\gamma L)]$

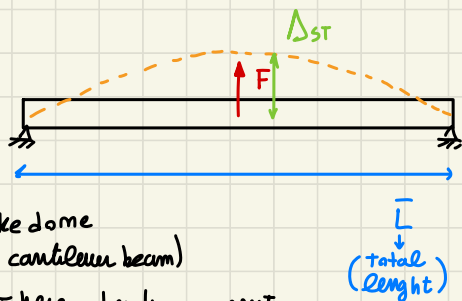
$$[H(\gamma L)] = \begin{bmatrix} \dots & \dots & \dots \\ \sin(\gamma L) & -\cos(\gamma L) & \sinh(\gamma L) & \cosh(\gamma L) & -C_c & 1 & -C_c & -1 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

$[H(\gamma L)](5;:)$

↓ implemented on **MATLAB**

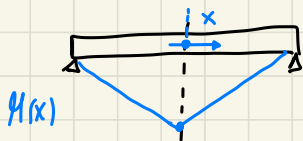
firstly we

found a reasonable value of **K** to use:



(like dome on cantilever beam)

BUT here bending moment will be linear as



discontinuity of $M(x)$ ↓

take 2 sub domains of integration for M

When removing K ,

simply supported beam

K reasonable =

stiffness when we apply F

static at center, compute

Δ_{st} of central section

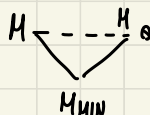
and $K = F / \Delta_{st}$

$$\frac{\partial^2 W}{\partial x^2} = \frac{M}{EI}$$

↓ integrating 2 times ... to set

$$C_1 = C_2 = 0$$

integr constant



$M(x)$ STATIC BENDING moment → change across beam length

$\Rightarrow$ you find so that $\Delta_{ST} = \frac{F \bar{L}^3}{EI 48} = \frac{8 F L^3}{EI 48} = \frac{F L^3}{6 EI}$
 $W(L) = \frac{F L^3}{6 EI}$
 $\bar{L} = 2L$

$K = \frac{F}{\frac{F L^3}{6 EI}} = \frac{6 EI}{L^3}$

MATLAB

name func = 'cont-beam-elastic-support'

$K = 6 * EI / L^3$; as found from calculation

global (K) (course inside the function)
 L LOW STIFF

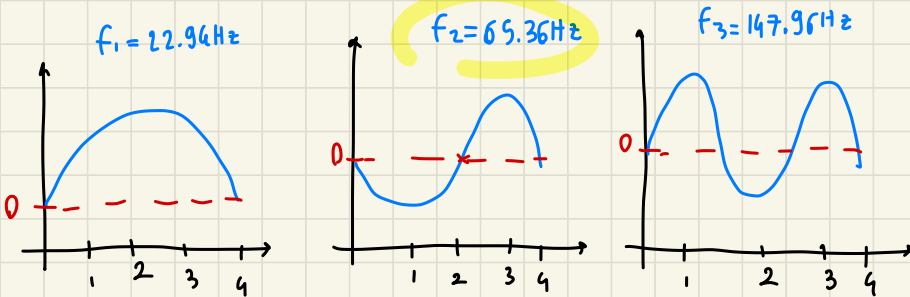
$f_{MAX} = 200 \text{ Hz}$ } 3 nat freq

here I define

$\gamma = \text{gamma} L / L$

$cc = K / EI / L^3$

↓
 and write [H] = ...



still STRUCTURAL SPM so we get symm, anti-symm shape

the anti symm has 0 displacement @ $L/2$ so @ f_2
 the spy does not affect $\Rightarrow$ get same result
 of PIN middle

While symm f and nat

freq we have max displacement on center $\Rightarrow$ so small k

don't keep beam fixed

if $K \uparrow \Rightarrow$ new $K' \approx 10^4 K$

we get f_i, ϕ_i as the same of ESI case

alternatively sPM/anti-sPM

$\approx K$ don't affect anti symm modal
 shapes here we have modal points

if $k=0 \rightarrow$ simply supported beam with doubled length
respect previous case

$\downarrow$ so f_k, ϕ_k are same as simply supp-beam